MIL-S-36500A
21 November 1969
SUPERSEDING
MIL-S-36500 dated
8 August 1966

MILITARY SPECIFICATION

STERILIZER, SURGICAL INSTRUMENT AND DRESSING, PRESSURE, EXTERNALLY HEATED

This specification is mandatory for use by all departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of portable, vertical, pressure sterilizer, externally heated and suitable for sterilizing surgical instruments and dressings in the field.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-640	Box, Fiberboard, Corrugated, Triple Wall.

FSC 6530

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:42Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center

24020	Autoclave, Lightweight, Field, and Bill of Material.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publication. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society of Mechanical Engineers (ASME)

POWER AND PRESSURE VESSEL CODE, SECTION I

(Applications for copies should be addressed to American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified in DPSC Drawing 24020 and Bill of Material thereto. Material gages specified shall be minimum. Commercial tolerances shall apply unless otherwise indicated.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:42Z
Check the source to verify that this is the current version before use.

* 3.2 Construction. The sterilizer shall consist of a sterilizing chamber with removable rack and trays, a cover, a water reservoir (steam generating chamber), two safety valves, a safety release and two control valves. The sterilizer shall be in strict accordance with DPSC Drawing 24020 and Bill of Materials thereto. The sterilizer shall withstand the hydrostatic pressure test specified in 4.4.1.2 without leakage, permanent set or deformation.

3.2.1 Components.

3.2.1.1 Safety valves. The sterilizer shall be equipped with two safety valves, of the weighted gravity type. One valve shall vent the pressure in the steam generator at 30 psig $\pm$ 2 and the other valve shall vent the pressure in the sterilizing chamber at 18 psig $\pm$ 2, when tested as specified in 4.4.1.2.

* 3.2.1.2 Safety release. The sterilizer shall have a safety release in the cover. The safety release shall vent the pressure in the sterilizing chamber, if it exceeds 25 psig $\pm$ 5. The safety release shall be tested as specified in 4.4.1.3.

3.2.1.3 Valves. The sterilizer shall have a valve to control the steam flow from the generator to the chamber and an over-flow valve to limit the water level in the generator. The valves shall show no evidence of leakage when tested in accordance with 4.4.1.2.

3.2.1.4 Legs. The sterilizer shall have three removable legs which are capable of being fitted into the sterilizing chamber for storage or shipment.

3.2.1.5 Removable rack. A removable rack having a fixed bottom shelf, two upper adjustable shelves and an adjustable top condensation guard shall be provided for use in the sterilizing chamber.

* 3.2.2 Welding. All welding of corrosion-resisting steel shall be by the inert gas method. Complete penetration of welds shall be accomplished to provide a strong weld. Where welding rod is used, it shall be 316L and the resultant weld shall be free of pits, discontinuities, flux, scale and laps.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:42Z
Check the source to verify that this is the current version before use.

* 3.2.2.1 Passivation of welds. All welds shall be passivated after fabrication, as a final treatment, by complete immersion in an acid bath at a temperature of 120° F. to 140° F. for a period of 15 to 30 minutes. The acid bath shall consist of 6 to 15 percent by volume (100 to 200 grams per liter) of nitric acid (70%) and 1/2 to 1-1/2% by volume (6 to 8 grams per liter) of hydrofluoric acid (60%). After this treatment, all components shall be washed thoroughly with clear, hot water until all traces of chemicals have been removed. Passivation shall not be required if Heliarc welding is used exclusively.

3.3 Performance. The sterilizer shall be capable of accomplishing 100% kill (as indicated by the melting of the sterilization indicator) at 250° F., when tested as specified in 4.4.1.1. The temperature shall not fall below 250° F. during the sterilization cycle. The temperature in the sterilizing chamber shall be controlled by the weighted gravity relief valve that is set to vent steam at 18 psig. As the valve vents the steam, air and condensate are driven out of the chamber. Dressings shall be dry and unscorched at the end of the test cycle.

3.3.1 Pressure cylinder. Each chamber of the pressure cylinder shall withstand, without leakage or deformation, a hydrostatic pressure equal to three times its working pressure (54 psig for the chamber and 90 psig for the generator) in accordance with the ASME Power and Pressure Vessel Code, Section I. The pressure cylinder shall be tested as specified in 4.4.1.2.

3.4 Repair parts. The following repair parts shall be provided with each sterilizer:

One cover gasket

One safety plug for the cover.

* 3.5 Finish. All exterior and interior corrosion-resisting steel surfaces shall have a smooth No. 2B mill finish.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:42Z
Check the source to verify that this is the current version before use.

3.6 Service data. The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. The service data shall include detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusinn as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the contracting officer. The preliminary service data shall be submitted sufficiently early to permit adequate review (based on a miximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. One set of the service data, as indicated above, shall be placed in a sealed, waterproof envelope, and affixed securely to the outside of the exterior container. The envelope shall be secured to the exterior container by use of metal ties, nails, or similar means, and shall be located at the end of the exterior container, preferably, when space permits, at the end bearing the markings required by MIL-STD-129. In cases where complete service data is too bulky to be conveniently attached to the outside of the exterior container, the contents of the exterior envelope may be limited to information necessary for roughing-in preparations, and a note shall be made on the exterior envelope which shall indicate where the remainder of the first set of service data will be located.

When two or more containers are required for packing a disassembled item (components of a single stock number), set markings shall be applied in accordance with MIL-STD-129. The envelope shall be affixed to the container designated as No. 1 of the set. The second set and, if necessary, the remainder of the first set of service data, shall be attached to the unit inside the exterior container. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:42Z
Check the source to verify that this is the current version before use.

Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.7 Identification marking. The sterilizer shall have a nameplate attached by means of rivets or drive screws. The nameplate shall be permanently marked with the following information:

Manufacturer's name
Model number
Serial number
Contract or purchase order number
Federal stock number
U.S.

Nameplates secured by adhesives shall not be used.

* 3.8 Instruction plate. The sterilizer shall have a metal plate permanently attached to the unit, with legible operating instructions printed permanently thereon as follows:

1. Open left valve (move handle to upward position) and right valve (move handle toward side of generator).

2. Remove contents from chamber and pour one gallon of water into chamber.

3. When water pours through left valve, close both valves and empty excess water from chamber.

4. Place items to be sterilized on racks, place cover over racks and insert in chamber.

5. Replace chamber lid and lock tightly into position.

6. Place sterilizer over heat source.

7. When generator relief valve (to left of left hand valve) pops off, open right valve with stick or rod.

8. After 30 minutes sterilization cycle, remove heat source, relieve pressure in chamber by lifting weight of chamber relief valve (right of right valve) with stick, open lid, remove items.

NOTE: A. Rack cover must be in place to insure dry packs.

B. Refill with water prior to each sterilization cycle.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:42Z
Check the source to verify that this is the current version before use.

* 3.9 Workmanship. Workmanship shall be first class throughout. The sterilizer shall be free from defects which detract from its appearance or may impair its serviceability. The sterilizer and all its accessories shall be free from sharp edges. Discoloration of steel and brass parts due to welding and brazing, shall not be cause for rejection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

Table I. Sampling for Examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	I	1.0
Total (major and minor defects)	I	4.0
Dimensional defects	S-3	2.5

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:42Z
Check the source to verify that this is the currentversion before use.

* 4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as specified below:

Table II. Sampling for Tests

Component (Unit of Product)	Characteristic	Requirement	Test Prodedure	Inspection level
Sterilizer	Hydrostatic Pressure	3.2	4.4.1.2	S-3
Valves	Performance	3.2.1.1	4.4.1.2	S-3
Safety Release	Performance	3.2.1.2	4.4.1.3	S-3
Valves	Performance	3.2.1.3	4.4.1.2	S-3
Sterilizer	Performance	3.3	4.4.1.1	S-1
Pressure Cylinder	Performance	3.3.1	4.4.1.2	S-3

4.3 Examination. The sterilizer shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects.

Table III. Categories and defects*

Major

101 Any component not functioning properly.
102 Any component missing, broken, or otherwise unusable.
103 Exposed edges not free of burrs and sharpness.
104 Threaded parts stripped, or do not function smoothly.
105 Pipe and pipe connections leak, not securely mounted or not neatly assembled.
106 Identification plate missing, incomplete, or illegible.
107 Instruction plate missing, incomplete, illegible or not securely attached.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:42Z
Check the source to verify that this is the current version before use.

Categories and defects* (Con't)

Major
108 Exposed welds not ground smooth to blend with adjacent surfaces.
109 Service data missing or incomplete.

Minor
201 Sterilizer does not rest evenly on flat surface.
202 Finish scratched, marred or otherwise defective.
203 Shelf insert does not rest on bottom of chamber without rock or wobble.

*Examination shall not be restricted to the classified possible defects listed.

4.3.2 Dimensional examination. The sterilizer shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classifed as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Performance.

4.4.1.1 Temperature. The sterilizer shall be operated in accordance with the operating instructions. Accurately calibrated thermocouples and maximum reading thermometers shall be used to determine the temperature attained in the sterilizing chamber throughout a complete sterilization cycle. Sterilization indicators (tablet-in-tube type), and suitable dressing pack shall be used in the chamber during the sterilizing cycle. The test report shall indicate the temperature range after a 250° F. temperature is reached in the chamber, the maximum temperature, and whether the sterilization indicators melted during the test.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:42Z
Check the source to verify that this is the currentversion before use.

4.4.1.2 Hydrostatic pressure. Three pressure relief valves of both the 30 psig and the 18 psig sizes shall be tested. Each sample shall be connected to a steam source. The report shall include the psig at which leakage occurs on each sample. The steam generator and the sterilizing chamber shall be connected by means of suitable fittings and tubing to a hydrostatic pressure source. The unused parts of the sterilizer shall be blocked by plugs. The chamber cover shall be closed and tightened to the fully closed position. The sterilizing chamber and steam generator shall be filled with water. The chamber shall be subjected to a pressure of 54 psig for a period of 15 minutes. The steam generator shall be subjected to a pressure of 90 psig for a period of 15 minutes.

* 4.4.1.3 Safety release. The sterilizing chamber shall be connected by means of suitable fittings and tubing to a steam source. The unused parts of the sterilizer shall be blocked by plugs. The sterilizing chamber shall be subjected to a steadily increasing pressure up to 25 psig or until the safety release blows.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One sterilizer, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. The removable rock with adjustable shelves, the three removed legs, and repair parts shall be wrapped with flexible corrugated paperboard and cushioned within the sterilizer in such a manner, as to prevent shifting of or damage to the contents. The sterilizer, with cover secured, shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Carton shall be designed with interior fittings to insure against shifting of or damage to the sterilizer. Closure shall be as specified in the appendix of PPP-B-636.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C as specified, (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:42Z
Check the source to verify that this is the current version before use.

5.2.1 Level A. Each unit shall be overboxed in an exterior container designed for a type I load and constructed in accordance with PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-640, class 2, style E. Closure and strapping shall be as specified in the applicable box specification.

5.2.2 Level B. Each unit shall be overboxed in an exterior container designed for a type I load and constructed in accordance with PPP-B-601, domestic type, PPP-B-621, class 1, or PPP-B-640, class 2, style E. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked to identify contents and quantity contained.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The sterilizer described herein is intended for use by Special Warfare personnel for sterilization of surgical instruments and dressings in remote areas under rugged field conditions. For use with hot plate, gasoline burner, or open fire.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable levels of packaging and packing (See 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:42Z
Check the source to verify that this is the currentversion before use.

6.3 This specification does not include all types, classes, grades, and sizes of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, which are normally procured to meet Military requirements.

6.4 This specification covers FSN 6530-926-2022, Sterilizer, Surgical Instrument and Dressing, Pressure, Externally Heated, appearing in the Federal Supply Catalog.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:42Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4956 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
Attn: DPSC-RSTS
2800 South 20th Street
Philadelphia, PA 19101

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-36500A Notice 2	STERILIZER, SURGICAL INSTRUMENT AND DRESSING, PRESSURE, EXTERNALLY HEATED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:42Z
Check the source to verify that this is the current version before use.